3) CS-stage with triode load :-

→ A MOS device operating in deep triode region behaves as a resistor and can therefore serve as the load in a CS stage.

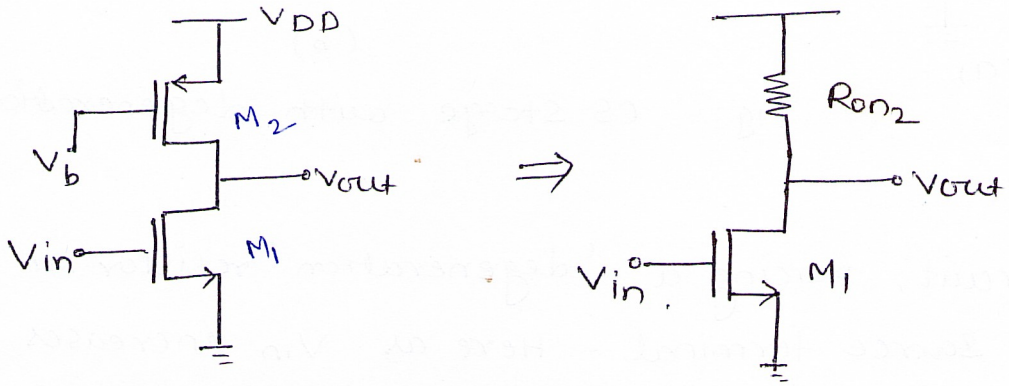


Fig: (a) CS-stage with triode load

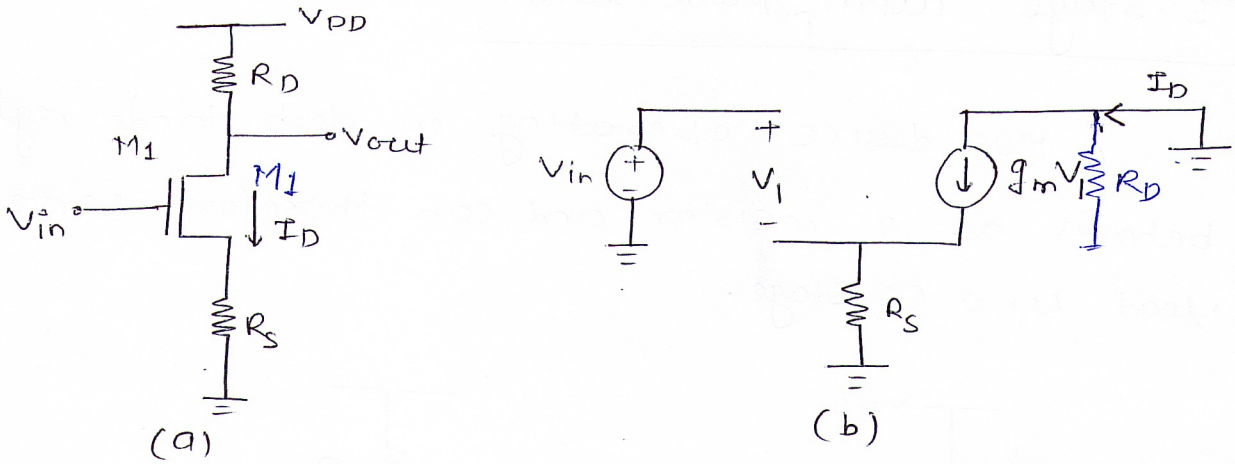
→ In above circuit, circuit biased gate of M_2 at a sufficient low level, ensuring load is in deep triode region for all o/p voltage swing.

Since,
$$R_{on_2} = \frac{1}{\mu_p C_{ox} \left(\frac{W}{L}\right)_2 (V_{DD} - V_b - V_{THp})}$$

we can calculate voltage gain.

disadvantage of above circuit is dependence of R_{on_2} upon $\mu_p C_{ox}$, V_b and V_{THp}. Since $\mu_p C_{ox}$ and V_{THp} vary with process and temperature and since generating precise value for V_b require addition complexity this circuit is difficult to use.

CS stage with Source Degeneration:-



→ Fig :- CS stage with degeneration

In above circuit, placing a 'degeneration' resistor in series with the source terminal. Here as Vin increases, so do I_D and the voltage drop across R_S.

i.e. a fraction of Vin appears across the resistor ~~acro~~ rather than as the gate-source overdrive, thus leading to a smoother variation of I_D.

Since $V_{out} = V_{DD} - I_D R_D$, ~~$V_{out} = -I_D R_D$~~

the non-linearity of the circuit arises from the non-linear depen-dence of I_D upon Vin.

$$\frac{\partial V_{out}}{\partial V_{in}} = -\left(\frac{\partial I_D}{\partial V_{in}}\right) \cdot R_D \qquad \text{---} \quad (21)$$

equivalut transconductance of the circuit,

$$G_m = \frac{\partial I_D}{\partial V_{in}}$$

assume, $I_D = f(V_{GS})$

$$\therefore \quad G_m = \frac{\partial I_D}{\partial V_{in}} = \frac{\partial I_D}{\partial V_{GS}} \times \frac{\partial V_{GS}}{\partial V_{in}} \qquad - (22)$$

$\therefore$ $V_{GS} = V_{in} - I_D R_S$ (from above circuit diagram)

$$\frac{\partial V_{GS}}{\partial V_{in}} = 1 - R_S \frac{\partial I_D}{\partial V_{in}}$$

from eqn (22)

$\therefore$ $$G_M = \left(1 - R_S \frac{\partial I_D}{\partial V_{in}}\right) \cdot \frac{\partial f}{\partial V_{GS}}$$

$g_m = \frac{\partial f}{\partial V_{GS}}$ $\longrightarrow$ transconductance of M_1

$$\therefore \quad G_M = \left(1 - R_S \cdot G_M\right) \cdot g_m$$

$$G_M = \frac{g_m}{1 + g_m R_S} \qquad \text{---} \quad (23)$$

$\therefore$ we know small-signal voltage gain, $A_V = -G_M R_D$

$\therefore$ without consideration any non-ideal effect

$$\boxed{A_V = -\left(\frac{g_m}{1 + g_m R_S}\right) \cdot R_D} \quad - (24)$$

from eq (23) as $R_S \uparrow$ es $\Rightarrow$ G_M becomes weaker function of g_m and hence drain current.

for

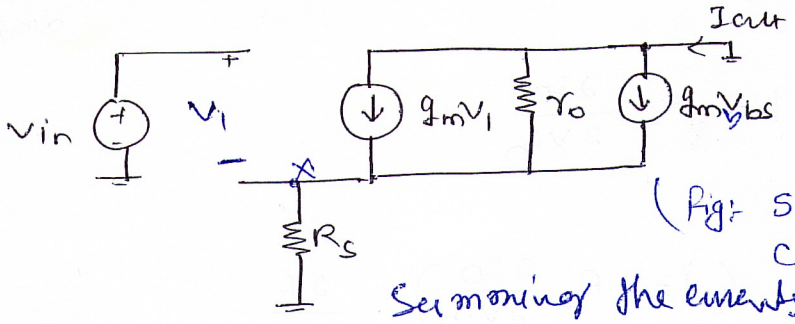
if, $R_S >> 1/g_m$ $\Rightarrow$ $G_m \simeq \frac{1}{R_S}$

$\Rightarrow$ $\Delta I_D \simeq \frac{\Delta V_{in}}{R_S}$

$\Rightarrow$ most of the change in V_{in} appears across R_S.

$\Rightarrow$ drain current is a linearized func. of i/p Vtg. but it cost of lowering the gain & higher noise.

now consider substrate bias effect, (body effect) + CLM, + The current through R_S equals I_{out} and therefore $V_{in} = V_1 + I_{out} R_S$.



(fig: small signal equivalent ckt of a degenerated CS stage

Summing the currents at node X, we have

from ckt, $I_{out} = +g_m V_1 - g_{mb} V_x + \frac{(-I_{out} R_S)}{r_o}$ (V_x)

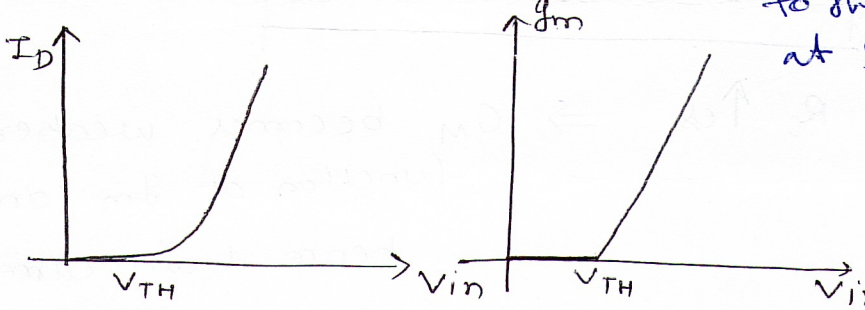
$\because V_{in} = V_1 + I_{out} R_S$

$\therefore I_{out} = g_m(V_{in} - I_{out} R_S) + g_{mb}(-I_{out} R_S) + \frac{(-I_{out} R_S)}{r_o}$ (V_x)

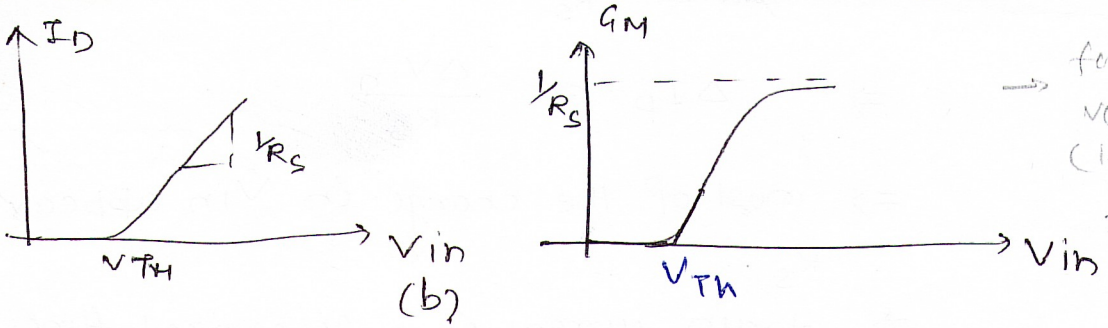
$G_M = \frac{I_{out}}{V_{in}}$ — (25)

$\therefore G_M = \frac{g_m r_o}{R_S + [1 + [g_m + g_{mb}]R_S] r_o}$ — (26)

Let us now examine the large-signal behavior of the CS stage if $R_S = 0$, indicate that I_D and g_m vary as shown in Fig. (a). For $R_S \neq 0$, the turn-on behaviour is similar to that in Fig (a) because at low current levels, $1/g_m >> R_S$ and hence $G_m \approx g_m$ (Fig. b). As the overdrive and g_m increase, the effect of degeneration



With, $R_S \neq 0$ (a)



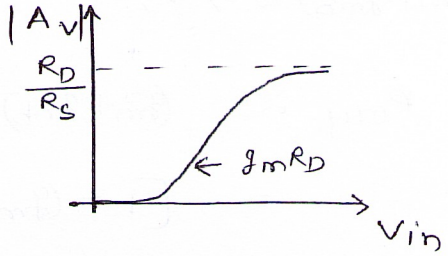
(b)

→ for large value of Vin (if M1 is still in saturation) ID is approximately linear and GM approaches 1/RS.

Fig:- drain current & transconductance of CS device
(a) without R_S
(b) with R_S

Note:- As Vin ↑es, degeneration becomes more significant and $A_v = -g_m R_D / 1 + g_m R_S$ ..

→ for large value of Vin, $G_M \approx \frac{1}{R_S}$ & $A_V = -\frac{R_D}{R_S}$.

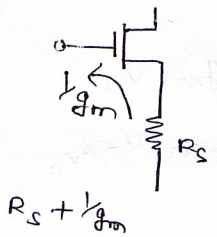


from eqn (23),

$$A_V = -\frac{R_D}{1/g_m + R_S} \qquad \text{—— (27)}$$

if Vin slightly greater than, V_{TH}, $1/g_m >> R_S \Rightarrow A_V \approx -g_m$

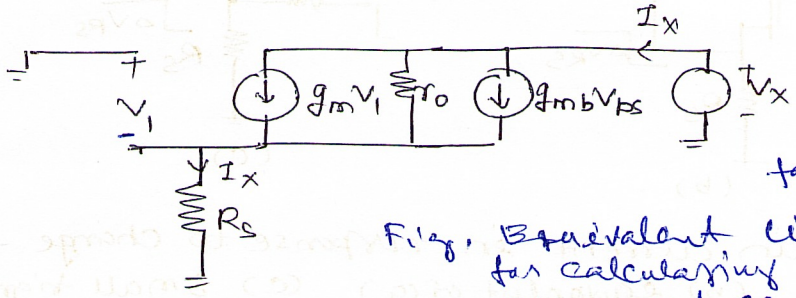
Note:-



$$|A_V| = \frac{\text{resistance seen at drain}}{\text{resistance seen in the source path}}$$

Fig:- Resistance seen ~~from~~ in source path

Calculation of o/p resistance:- (considering body effect)



Another important conse... of some degeneration is ... increase in the output resi... tance of the stage.

Fig. Equivalent circuit ~~flowing through~~ for calculating the o/p resistance of a de... generated CS stage.

from ckt, $V_1 = -I_x R_S$

current flowing through r_o, $I_x - (g_m + g_{mb}) V_1 = I_x + (g_m + g_{mb}) R_S I_x$

now, adding the vtg drop across r_o & R_S,

$$r_o [I_x + (g_m + g_{mb}) R_S I_x] + I_x R_S = V_x$$

$$\therefore \quad R_{out} = \frac{V_x}{I_x} = \left[1 + (g_m + g_{mb}) R_s\right] r_o + R_s \qquad -(28)$$

$$= \left[1 + (g_m + g_{mb}) r_o\right] R_s + r_o \qquad -(29)$$

since typically, $(g_m + g_{mb}) r_o \gg 1$,

$$R_{out} \approx (g_m + g_{mb}) r_o R_s + r_o \qquad (30)$$

$$= \left[1 + (g_m + g_{mb}) R_s\right] r_o \qquad -(31)$$

$\Rightarrow$ o/p resistance raised by factor $[1 + (g_m + g_{mb}) R_s]$

from above ckt, if $R_s = 0$, $g_m V_1 = g_{mb} V_{bs} = 0$

$$\therefore I_x = V_x / r_o$$

$R_s > 0 \Rightarrow I_x R_s > 0$ & $V_1 < 0$

get, -ive $g_m V_1$ & $g_{mb} V_{bs}$

$\Rightarrow$ current supplied by V_x is less than V_x / r_o

above o/p Resistence can also calculate using following ckt inspection,

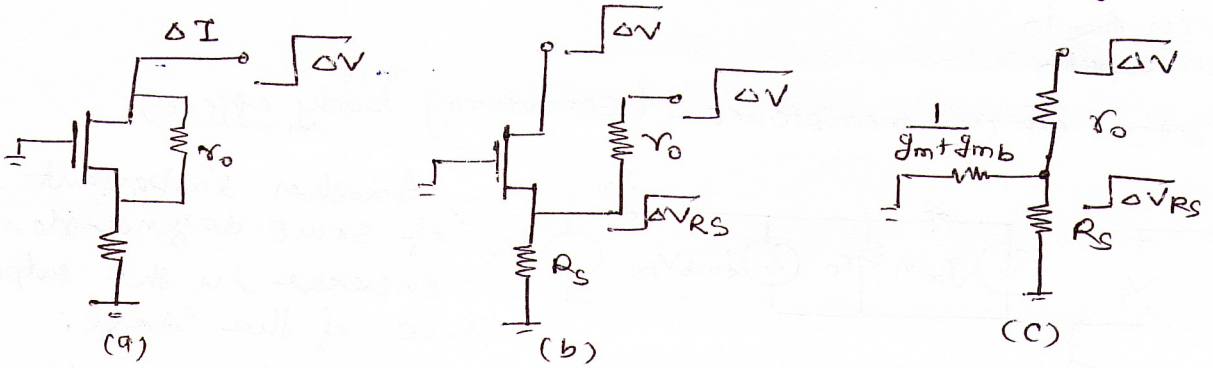


fig: (a) change in drain current in response to change in applied V/g gain (b) equivalent of (a) (c) small signal model

the V/g change across R_s, (by voltage devision)

$$\Delta V_{RS} = \Delta V \cdot \frac{\left(\frac{1}{g_m + g_{mb}}\right) \| R_s}{\frac{1}{(g_m + g_{mb})} \| R_s + r_o}$$

Change in current,

$$\Delta I = \frac{\Delta V_{RS}}{R_S}$$

$$= \Delta V \cdot \frac{1}{[1+(g_m+g_{mb})R_S]r_o + R_S}$$

$$\frac{\Delta V}{\Delta I} = [1+(g_m+g_{mb})R_S]r_o + R_S$$

same as eqn no. (20).

Calculate gain of a degenerated CS stage (considering body effect & channel length Modn)

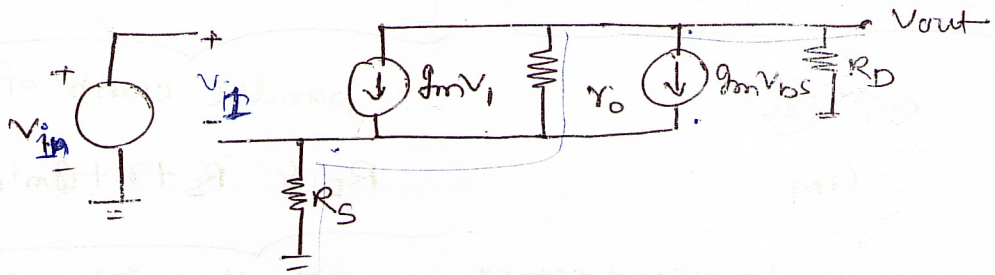


fig:- small signal model of degenerated CS stage with finite o/p resistance

from ckt,

$$V_1 = V_{in} + \frac{V_{out} R_S}{R_D} \qquad \because I_{R_S} = I_{R_D} = \frac{-V_{out}}{R_D}$$

$$\therefore I_{r_o} = -\frac{V_{out}}{R_D} - (g_m V_1 + g_{mb} V_{bs})$$

$$= -\frac{V_{out}}{R_D} - \left[g_m \left(V_{in} + V_{out}\frac{R_S}{R_D} \right) + g_{mb} V_{out} \frac{R_S}{R_D} \right]$$

Volt drop across r_o & R_S must add up to V_{out},

$\Rightarrow$

$$V_{out} = I_{r_o} r_o + \left(\frac{V_{out}}{R_D}\right) R_S$$

$$= -\frac{V_{out}}{R_D} r_o - \left[g_m \left(V_{in} + V_{out}\frac{R_S}{R_D} \right) + g_{mb} V_{out} \frac{R_S}{R_D} \right]$$

$$- V_{out} \cdot \frac{R_S}{R_D}$$

we can write,

$$A_V = \frac{V_{out}}{V_{in}} = \frac{-g_m r_o R_D}{R_D + R_S + r_o + (g_m + g_{mb}) R_S r_o} \quad -(32)$$

above expression can also written as, $(A_v = -g_m R_{out})$

→ $R_S + r_o + (g_m + g_{mb}) R_S r_o$ → represent o/p res. of mos device.

$$A_V = \frac{-g_m r_o R_D \left[R_S + r_o + (g_m + g_{mb}) R_S r_o\right]}{R_D + R_S + r_o + (g_m + g_{mb}) R_S r_o} \cdot \frac{1}{R_S + r_o + (g_m + g_{mb}) R_S r_o}$$

$$A_V = \underbrace{\frac{-g_m r_o}{\left(R_S + r_o + (g_m + g_{mb}) R_S r_o\right)}}_{\text{eqn (26)}} \underbrace{\left[\frac{R_D \left[R_S + r_o + (g_m + g_{mb}) R_S r_o\right]}{R_D + R_S + r_o + (g_m + g_{mb}) R_S r_o}\right]}_{\text{parallel comb}^n \text{ of}^2} \quad -(33)$$

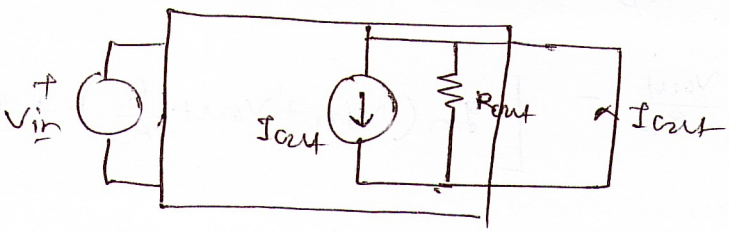
G_M — overall transconductance of ckt

R_D & $R_S + r_o + (g_m + g_{mb}) r_o R_S$ — overall resistance of ckt

Note: In a linear ckt, the voltage gain is equal to $-G_M R_{out}$, where G_M denotes transconductance of circuit when o/p is shorted to ground

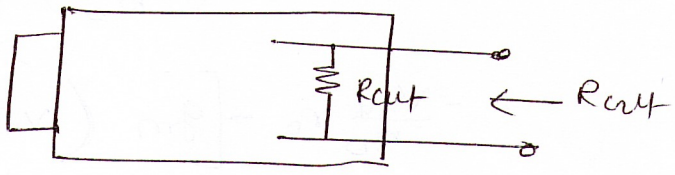
i.e.

Norton equivalent



$$G_M = \frac{I_{out}}{V_{in}}$$

R_{out} represent the o/p resistance of ckt when, i/p V+g set to zero.



$$V_{out} = -G_M V_{in} R_{out}$$